

Question	Answer	Mark	Mark scheme	Additional guidance
23	098.6	P1	for using bearings to determine ABC as 67°	Accept 67 written on the diagram.
		P1	for using the cosine rule to find AC eg $(AC^2 =) 9^2 + 8^2 - 2 \times 9 \times 8 \times \cos[67]$ oe or $AC = 9.4199\dots$	Accept correct substitution into RHS of equation Accept AC in the range 9.41 to 9.42
		P1	(dep P1) for using the sine rule to find angle BAC eg $\frac{9}{\sin BAC} = \frac{"9.42"}{\sin[67]}$ oe	
			OR	
			for using the cosine rule to find angle BAC eg $9^2 = "9.42^2" + 8^2 - 2 \times "9.42" \times 8 \times \cos BAC$ oe	
		P1	for rearranging eg $\sin BAC = 9 \times \frac{\sin[67]}{"9.42"}$ oe	Accept any equivalent form with values substituted
			OR	
			eg $\cos BAC = ("9.42^2" + 8^2 - 9^2) \div (2 \times "9.42" \times 8)$ oe	
			OR	
			for angle $BAC = 61.57\dots$	
		A1	for angle in the range 98.5 to 98.6	If the correct answer is given without supportive evidence award 0 marks. Condone missing "0" at the front. If an answer within the range is seen in working and rounded incorrectly award full marks.